1) $f(x) = x^2$ $\qquad 0 \le x \le 1$

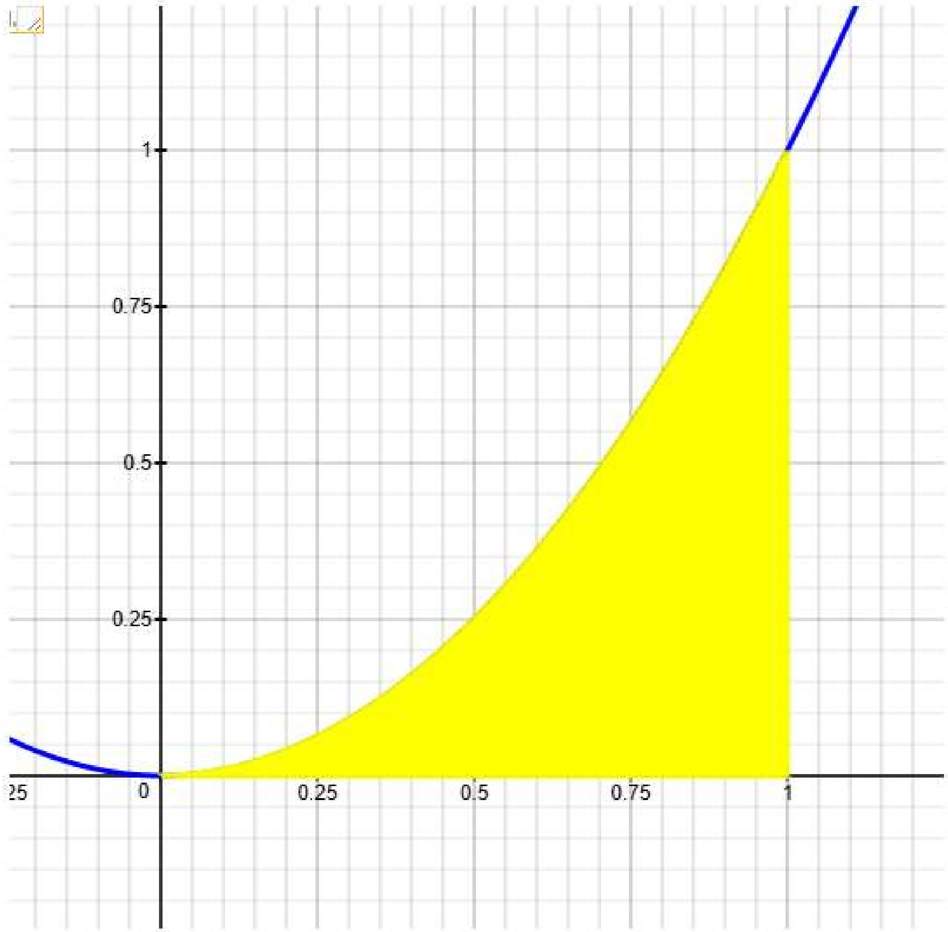
Find the shaded area between the x-axis and $f(x)$.

Let $a = 0$ and $b = 2$

Let $n =$ number of subintervals = 8.

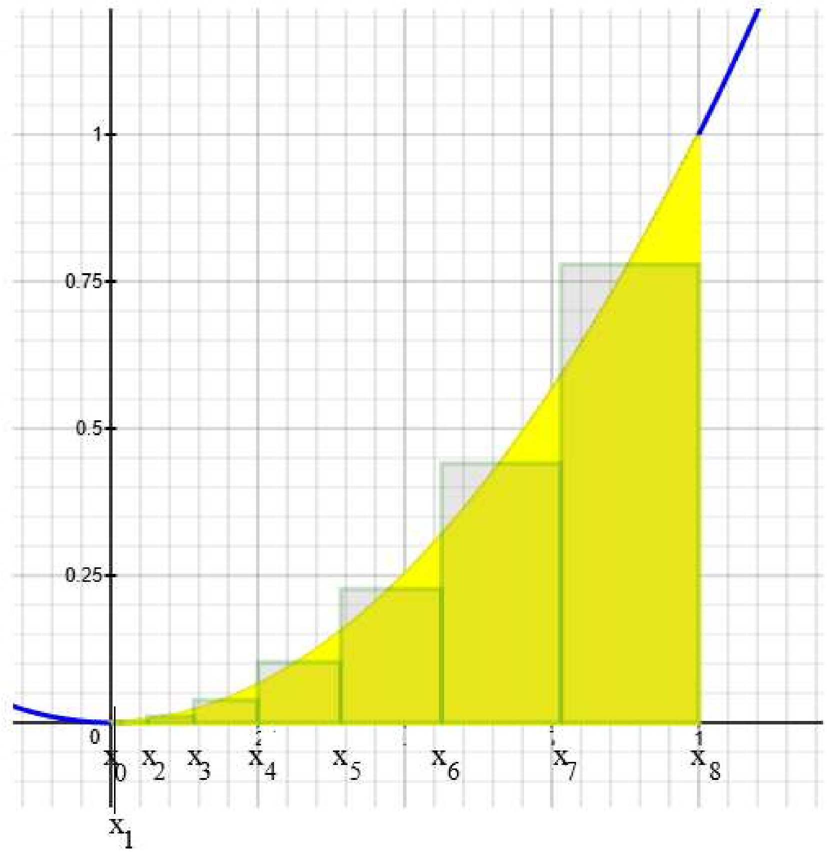
Partition the interval $[a, b]$ as follows:

$x_0 = 0$
$x_1 = 0.015625$
$x_2 = 0.0625$
$x_3 = 0.140625$
$x_4 = 0.25$
$x_5 = 0.390625$
$x_6 = 0.5625$
$x_7 = 0.765625$
$x_8 = 1$



Rectangle 1	$\Delta x_1 = x_1 - x_0 = 0.015625$ midpoint $= x_1^m = (x_1 + x_0)/2 = 0.0078125$ height $= f(0.0078125) = 0.00006103515625$ Area $= (0.00006103515625)(0.015625) = 9.53674\text{e-}7$
Rectangle 2	$\Delta x_2 = x_2 - x_1 = 0.046875$ midpoint $= x_2^m = (x_2 + x_1)/2 = 0.0390625$ height $= f(0.0390625) = 0.00152587890625$ Area $= (0.00152587890625)(0.046875) = 0.000071525574$
Rectangle 3	$\Delta x_3 = x_3 - x_2 = 0.078125$ midpoint $= x_3^m = (x_3 + x_2)/2 = 0.1015625$ height $= f(0.1015625) = 0.01031494140625$ Area $= (0.01031494140625)(0.078125) = 0.000805854797$
Rectangle 4	$\Delta x_4 = x_4 - x_3 = 0.109375$ midpoint $= x_4^m = (x_4 + x_3)/2 = 0.1953125$ height $= f(0.1953125) = 0.03814697265625$ Area $= (0.03814697265625)(0.109375) = 0.004172325134$
Rectangle 5	$\Delta x_5 = x_5 - x_4 = 0.140625$ midpoint $= x_5^m = (x_5 + x_4)/2 = 0.3203125$ height $= f(0.3203125) = 0.10260009765625$ Area $= (0.10260009765625)(0.140625) = 0.014428138733$

Note: $9.53674\text{e-}7 = 0.00000095367$



Rectangle 6	$\Delta x_6 = x_6 - x_5 = 0.171875$ midpoint $= x_6^m = (x_6 + x_5)/2 = 0.4765625$ height $= f(0.4765625) = 0.22711181640625$ Area $= (0.22711181640625)(0.171875)=0.039034843445$
Rectangle 7	$\Delta x_7 = x_7 - x_6 = 0.203125$ midpoint $= x_7^m = (x_7 + x_6)/2 = 0.6640625$ height $= f(0.6640625) = 0.44097900390625$ Area $= (0.44097900390625)(0.203125)=0.089573860168$
Rectangle 8	$\Delta x_8 = x_8 - x_7 = 0.234375$ midpoint $= x_8^m = (x_8 + x_7)/2 = 0.8828125$ height $= f(0.8828125) = 0.77935791015625$ Area $= (0.77935791015625)(0.234375)=0.182662010193$

a) Riemann Sum = Sum of areas of rectangles

$$= f\left(x_1^m\right) \cdot \Delta x_1 + f\left(x_2^m\right) \cdot \Delta x_2 + f\left(x_3^m\right) \cdot \Delta x_3 + f\left(x_4^m\right) \cdot \Delta x_4 + \ldots + f\left(x_8^m\right) \cdot \Delta x_8$$

$$= \sum_{i=1}^{8} f\left(x_i^m\right) \cdot \Delta x_i = \underline{\quad ? \quad}$$

b) $\|\Delta\| =$ norm of the partition = width of largest subinterval = $\underline{\quad ? \quad}$

c) If $\|\Delta\| \to 0$, what happens to the number of subintervals?

Meaning: If $\|\Delta\| \to 0$, then $n \to ?$

2) $f(x) = 0.5x^2 + 2 \qquad 0 \le x \le 1$

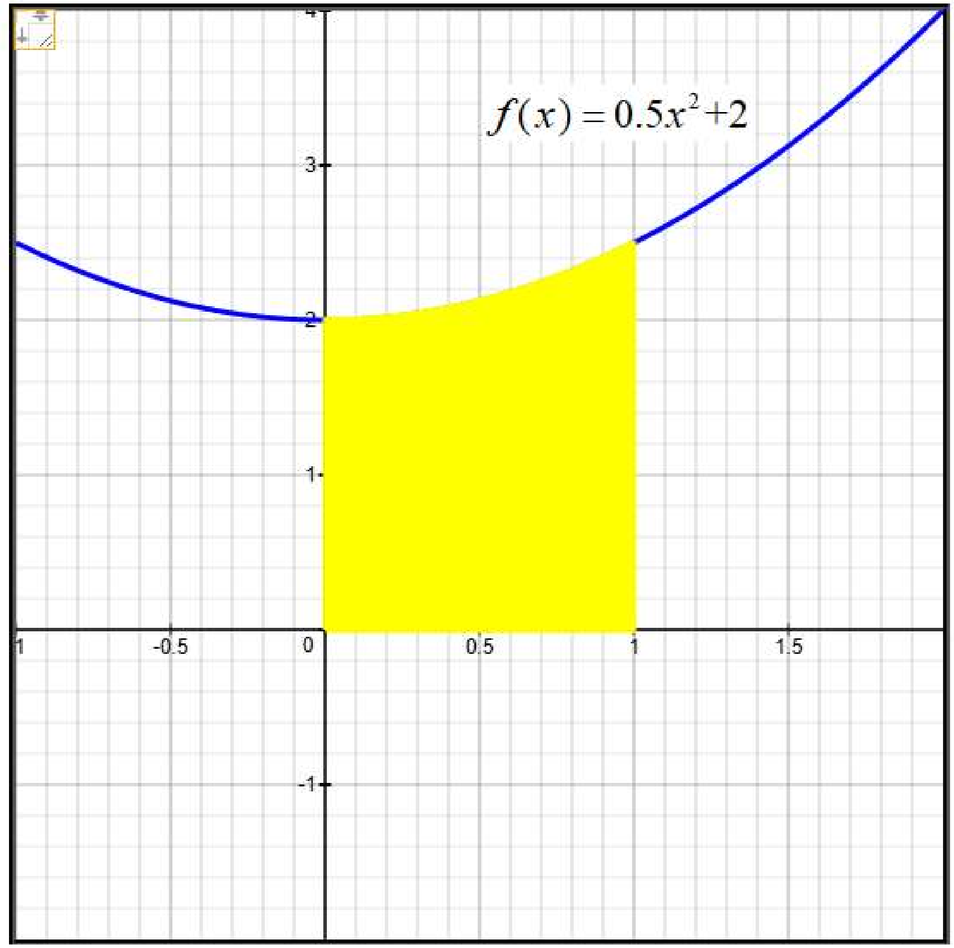
Find the shaded area between the x-axis and $f(x)$.

Let $a = 0$ and $b = 1$

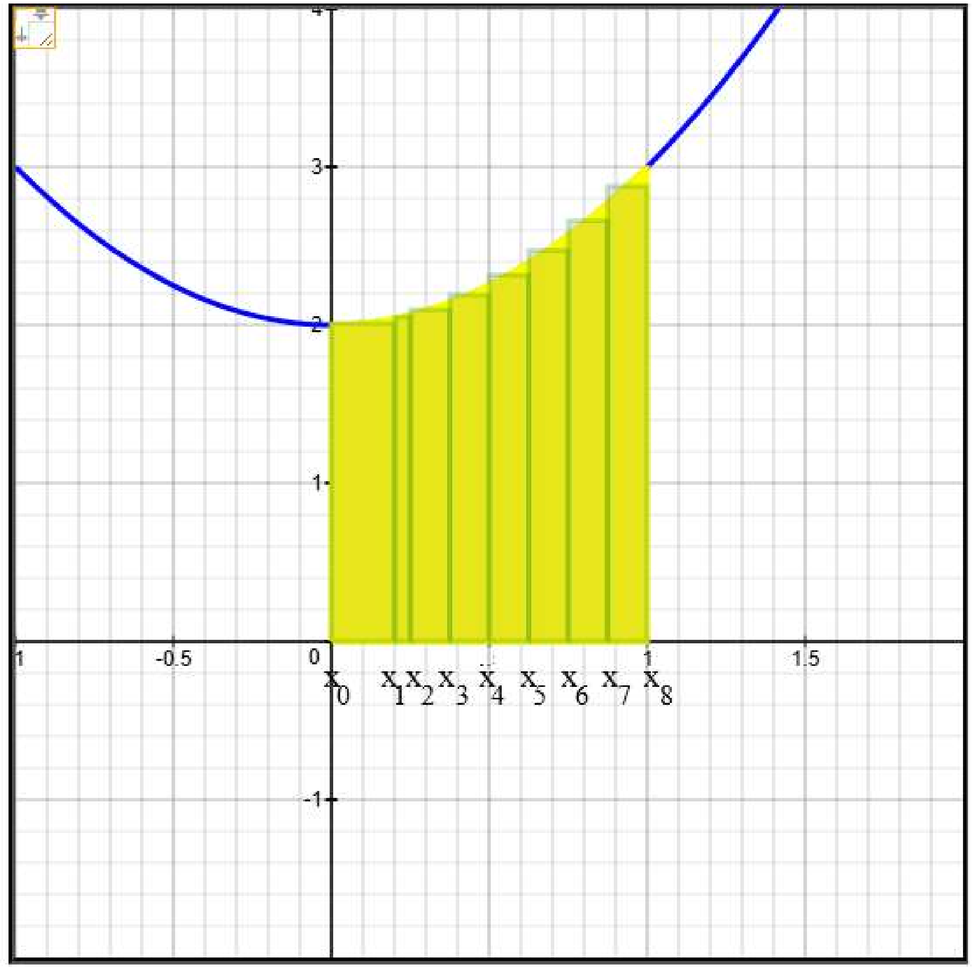
Let $n =$ number of subintervals = 8.

Partition the interval $[a, b]$ as follows:

$x_0 = 0$
$x_1 = 0.2$
$x_2 = 0.25$
$x_3 = 0.375$
$x_4 = 0.5$
$x_5 = 0.625$
$x_6 = 0.75$
$x_7 = 0.875$
$x_8 = 1$



Rectangle 1	$\Delta x_1 = x_1 - x_0 = 0.2$ midpoint $= x_1^m = (x_1 + x_0)/2 = 0.1$ height $= f(0.1) = 2.01$ Area $= (2.01)(0.2)=0.402$
Rectangle 2	$\Delta x_2 = x_2 - x_1 = 0.05$ midpoint $= x_2^m = (x_2 + x_1)/2 = 0.225$ height $= f(0.225) = 2.050625$ Area $= (2.050625)(0.05)=0.10253125$
Rectangle 3	$\Delta x_3 = x_3 - x_2 = 0.125$ midpoint $= x_3^m = (x_3 + x_2)/2 = 0.3125$ height $= f(0.3125) = 2.09765625$ Area $= (2.09765625)(0.125)=0.26220703125$
Rectangle 4	$\Delta x_4 = x_4 - x_3 = 0.125$ midpoint $= x_4^m = (x_4 + x_3)/2 = 0.4375$ height $= f(0.4375) = 2.19140625$ Area $= (2.19140625)(0.125)=0.27392578125$



Rectangle 5	$\Delta x_5 = x_5 - x_4 = 0.125$ midpoint = $x_5^m = (x_5 + x_4)/2 = 0.5625$ height = $f(0.5625) = 2.31640625$ Area = (2.31640625)(0.125)=0.28955078125
Rectangle 6	$\Delta x_6 = x_6 - x_5 = 0.125$ midpoint = $x_6^m = (x_6 + x_5)/2 = 0.6875$ height = $f(0.6875) = 2.47265625$ Area = (2.47265625)(0.125)=0.30908203125
Rectangle 7	$\Delta x_7 = x_7 - x_6 = 0.125$ midpoint = $x_7^m = (x_7 + x_6)/2 = 0.8125$ height = $f(0.8125) = 2.66015625$ Area = (2.66015625)(0.125)=0.33251953125
Rectangle 8	$\Delta x_8 = x_8 - x_7 = 0.125$ midpoint = $x_8^m = (x_8 + x_7)/2 = 0.9375$ height = $f(0.9375) = 2.87890625$ Area = (2.87890625)(0.125)=0.35986328125

a) Riemann Sum = Sum of areas of rectangles

$$= f\left(x_1^m\right) \cdot \Delta x_1 + f\left(x_2^m\right) \cdot \Delta x_2 + f\left(x_3^m\right) \cdot \Delta x_3 + f\left(x_4^m\right) \cdot \Delta x_4 + \ldots + f\left(x_8^m\right) \cdot \Delta x_8$$

$$= \sum_{i=1}^{8} f\left(x_i^m\right) \cdot \Delta x_i = \underline{\quad ? \quad}$$

b) $\|\Delta\| =$ norm of the partition = width of largest subinterval = $\underline{\quad ? \quad}$

c) If $\|\Delta\| \to 0$, what happens to the number of subintervals?

Meaning: If $\|\Delta\| \to 0$, then $n \to ?$

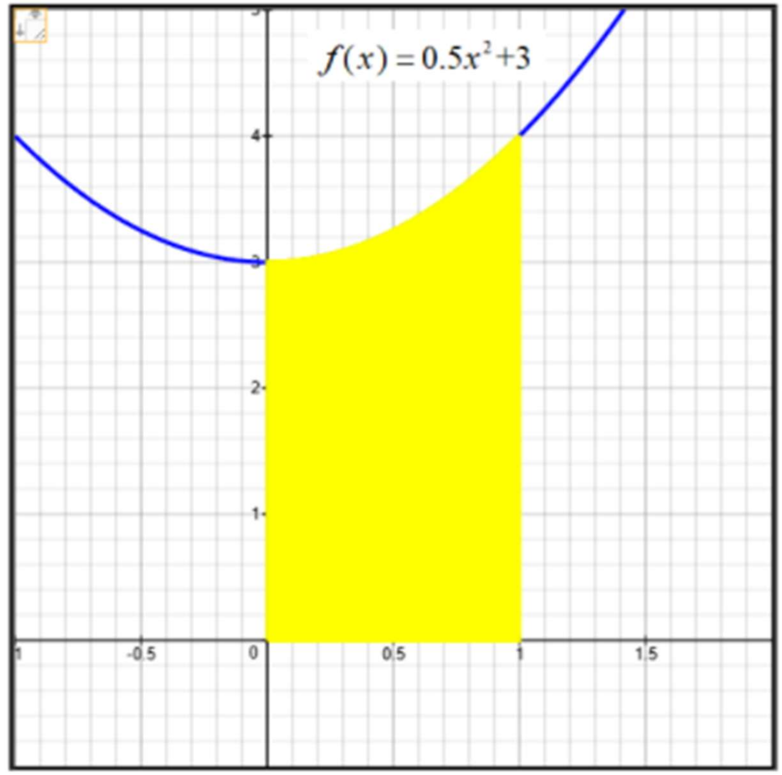
3) $f(x) = 0.5x^2 + 3 \qquad 0 \le x \le 1$

Find the shaded area between the x-axis and $f(x)$.

Let $a = 0$ and $b = 1$

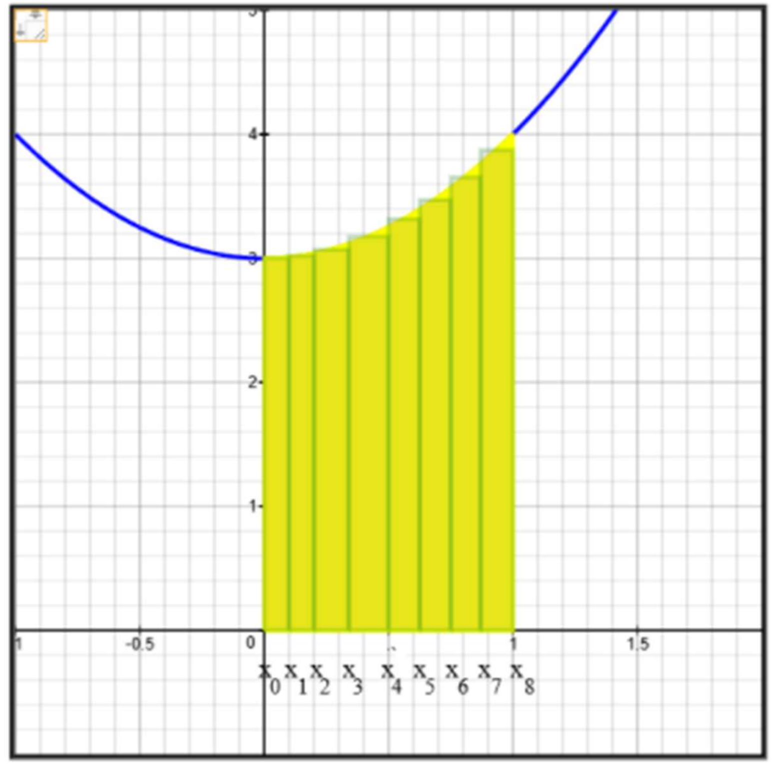
Let $n =$ number of subintervals = 8.

Partition the interval $[a, b]$ as follows:



$x_0 = 0$
$x_1 = 0.1$
$x_2 = 0.2$
$x_3 = 0.34$
$x_4 = 0.5$
$x_5 = 0.625$
$x_6 = 0.75$
$x_7 = 0.87$
$x_8 = 1$

Rectangle 1	$\Delta x_1 = x_1 - x_0 = 0.1$ midpoint = $x_1^m = (x_1 + x_0)/2 = 0.05$ height = $f(0.05) = 3.0025$ Area = $(3.0025)(0.1)=0.30025$
Rectangle 2	$\Delta x_2 = x_2 - x_1 = 0.1$ midpoint = $x_2^m = (x_2 + x_1)/2 = 0.15000000000000002$ height = $f(0.15000000000000002) = 3.0225$ Area = $(3.0225)(0.1)=0.30225$
Rectangle 3	$\Delta x_3 = x_3 - x_2 = 0.14$ midpoint = $x_3^m = (x_3 + x_2)/2 = 0.27$ height = $f(0.27) = 3.0729$ Area = $(3.0729)(0.14)=0.430206$
Rectangle 4	$\Delta x_4 = x_4 - x_3 = 0.16$ midpoint = $x_4^m = (x_4 + x_3)/2 = 0.42000000000000004$ height = $f(0.42000000000000004) = 3.1764$ Area = $(3.1764)(0.16)=0.508224$



Rectangle 5	$\Delta x_5 = x_5 - x_4 = 0.125$ midpoint $= x_5^m = (x_5 + x_4)/2 = 0.5625$ height $= f(0.5625) = 3.31640625$ Area = (3.31640625)(0.125)=0.41455078125
Rectangle 6	$\Delta x_6 = x_6 - x_5 = 0.125$ midpoint $= x_6^m = (x_6 + x_5)/2 = 0.6875$ height $= f(0.6875) = 3.47265625$ Area = (3.47265625)(0.125)=0.43408203125
Rectangle 7	$\Delta x_7 = x_7 - x_6 = 0.12$ midpoint $= x_7^m = (x_7 + x_6)/2 = 0.81$ height $= f(0.81) = 3.6561000000000003$ Area = (3.6561000000000003)(0.12)=0.438732
Rectangle 8	$\Delta x_8 = x_8 - x_7 = 0.13$ midpoint $= x_8^m = (x_8 + x_7)/2 = 0.935$ height $= f(0.935) = 3.874225$ Area = (3.874225)(0.13)=0.50364925

a) Riemann Sum = Sum of areas of rectangles

$$= f\left(x_1^m\right)\cdot\Delta x_1 + f\left(x_2^m\right)\cdot\Delta x_2 + f\left(x_3^m\right)\cdot\Delta x_3 + f\left(x_4^m\right)\cdot\Delta x_4 + \ldots + f\left(x_8^m\right)\cdot\Delta x_8$$

$$= \sum_{i=1}^{8} f\left(x_i^m\right)\cdot\Delta x_i = \underline{\quad ? \quad}$$

b) $\|\Delta\| =$ norm of the partition = width of largest subinterval = $\underline{\quad ? \quad}$

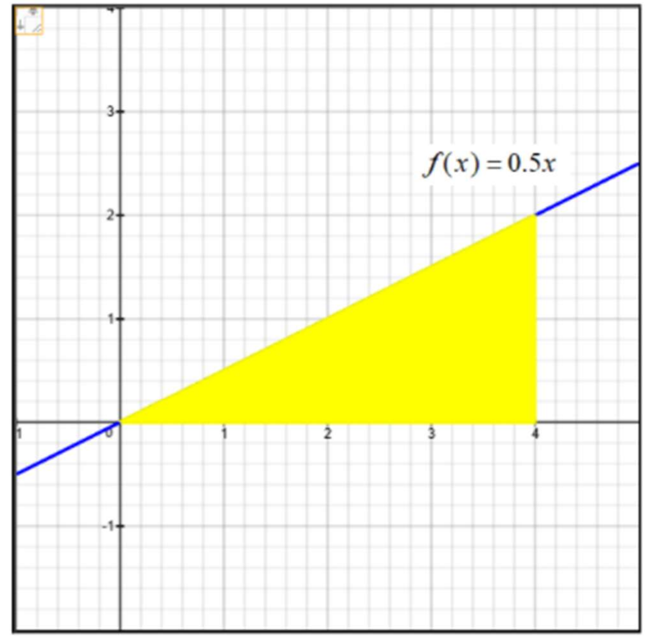
c) If $\|\Delta\| \to 0$, what happens to the number of subintervals?

Meaning: If $\|\Delta\| \to 0$, then $n \to ?$

4) $f(x) = 0.5x \qquad 0 \le x \le 4$

Find the shaded area between the x-axis and $f(x)$.

Let $a = 0$ and $b = 4$



Let $\|\Delta\| =$ norm of the partition = width of largest subinterval

Recall: If $\|\Delta\| \to 0$, then $n \to \infty$.

a) Area of Right Triangle = __?__

b) Area of shaded region = $\lim\limits_{\|\Delta\| \to 0} \sum\limits_{i=1}^{n} f(c_i) \cdot \Delta x_i$ = Area of Right Triangle = __?__

If the limit quantity $\lim\limits_{\|\Delta\| \to 0} \sum\limits_{i=1}^{n} f(c_i) \cdot \Delta x_i$ exists, it is defined as follows:

$$\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i) \cdot \Delta x_i = \int_a^b f(x) \cdot dx = \int_0^4 (0.5x) \cdot dx$$

5) $f(x)=\sqrt{9-x^2}$ $\quad\quad$ $-3 \le x \le 3$

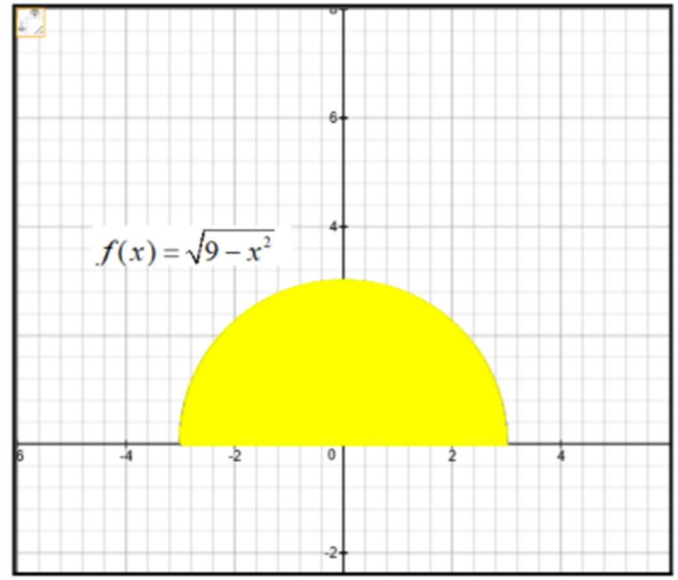
Find the shaded area between the x-axis and $f(x)$.

Let $a = -3$ and $b = 3$

Let $n =$ number of subintervals

Let $\|\Delta\| =$ norm of the partition = width of largest subinterval

Recall: If $\|\Delta\| \to 0$, then $n \to \infty$.



a) Area of Semicircle = __?__ $\quad\quad$ Hint: Area of circle = $\pi \cdot r^2$

b) Area of shaded region = $\lim\limits_{\|\Delta\|\to 0} \sum\limits_{i=1}^{n} f(c_i)\cdot \Delta x_i$ = Area of Semicircle = __?__

If the limit quantity $\lim\limits_{\|\Delta\|\to 0} \sum\limits_{i=1}^{n} f(c_i)\cdot \Delta x_i$ exists, it is defined as follows:

$$\lim_{\|\Delta\|\to 0} \sum_{i=1}^{n} f(c_i)\cdot \Delta x_i = \int_a^b f(x)\cdot dx = \int_{-3}^{3} \left(\sqrt{9-x^2}\right)\cdot dx$$

6) $f(x) = \text{-}2x + 4 \qquad \text{-}6 \le x \le 1$

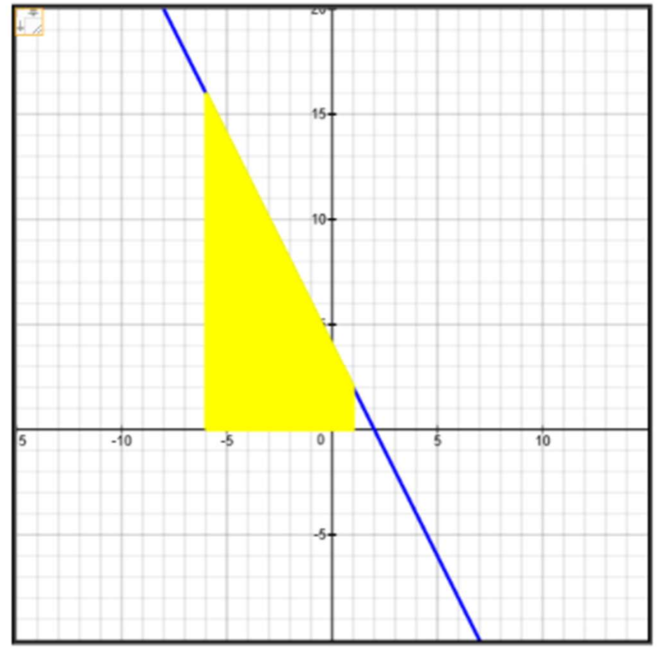
Find the shaded area between the x-axis and $f(x)$.

Let $a = \text{-}6$ and $b = 1$

Let $n =$ number of subintervals

Let $\|\Delta\| =$ norm of the partition = width of largest subinterval

Recall: If $\|\Delta\| \to 0$, then $n \to \infty$.



a) Area of Trapezoid = __?__ Hint: Area of trapezoid $= \frac{1}{2} \cdot h \cdot (b_1 + b_2)$

b) Area of shaded region = $\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i) \cdot \Delta x_i$ = Area of Trapezoid = __?__

If the limit quantity $\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i) \cdot \Delta x_i$ exists, it is defined as follows:

$$\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i) \cdot \Delta x_i = \int_a^b f(x) \cdot dx = \int_{-6}^{1} (\text{-}2x + 4) \cdot dx$$

7) $f(x) = 10 - |x| \qquad -3 \le x \le 3$

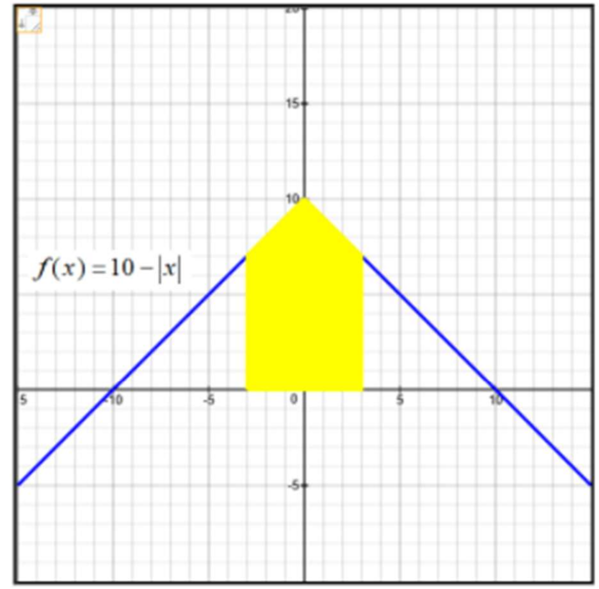
Find the shaded area between the x-axis and $f(x)$.

Let $a = -6$ and $b = 1$

Let $n =$ number of subintervals

Let $\|\Delta\| =$ norm of the partition = width of largest subinterval

Recall: If $\|\Delta\| \to 0$, then $n \to \infty$.

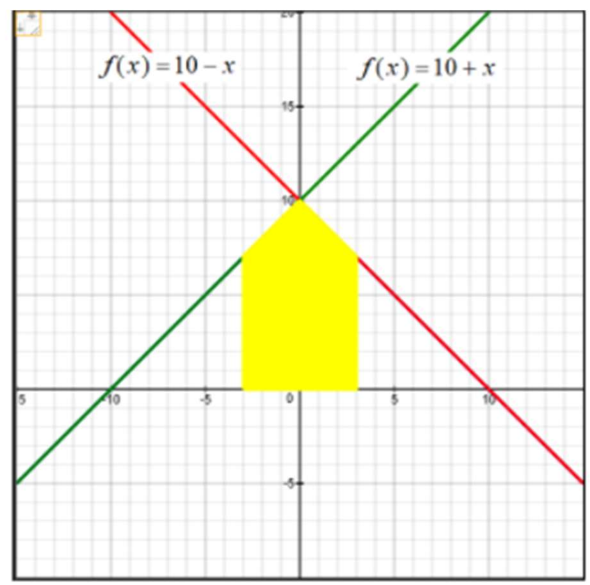


Note: $f(x) = 10 - |x| = \begin{cases} 10 - x & \text{if } x \ge 0 \\ 10 - -x = 10 + x & \text{if } x < 0 \end{cases}$

a) Area of Trapezoidal Region from $x = 0$ to $x = 3$ is __?__

b) Area of Entire Shaded Region from $x = -3$ to $x = 3$ is __?__

c) Area of shaded region = $\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i) \cdot \Delta x_i =$ __?__



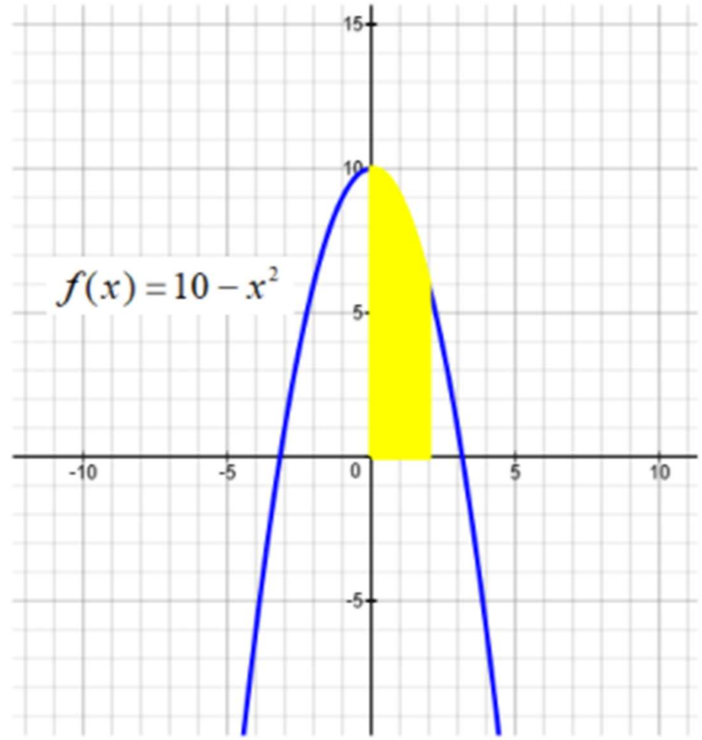
If the limit quantity $\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i) \cdot \Delta x_i$ exists, it is defined as follows:

$$\lim_{\|\Delta\| \to 0} \sum_{i=1}^{n} f(c_i) \cdot \Delta x_i = 2 \cdot \int_0^3 (10 - x) \cdot dx$$

8) $f(x) = 10 - x^2 \qquad 0 \le x \le 2$

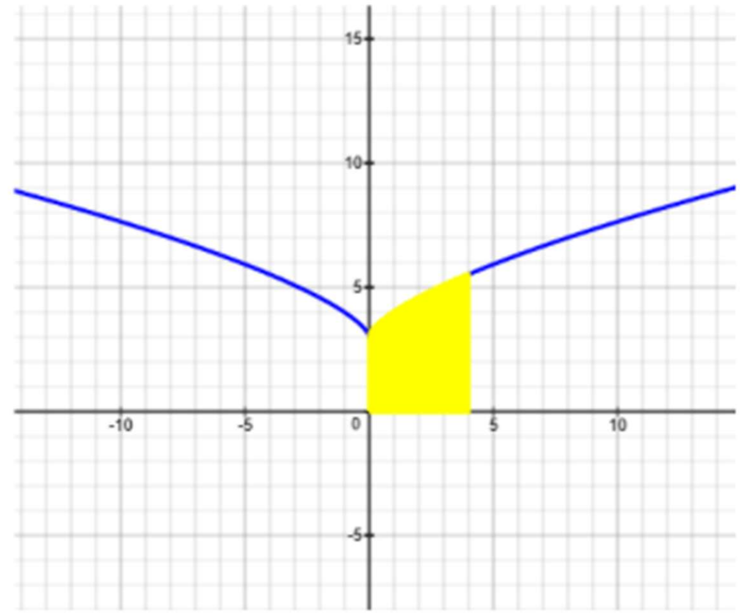
Set up the definite integral representing the shaded area.

Shaded Area = $\int_a^b f(x) \cdot dx =$ __?__



9) $f(x) = x^{2/3} + 3 \qquad 0 \le x \le 4$

Set up the definite integral representing the shaded area.

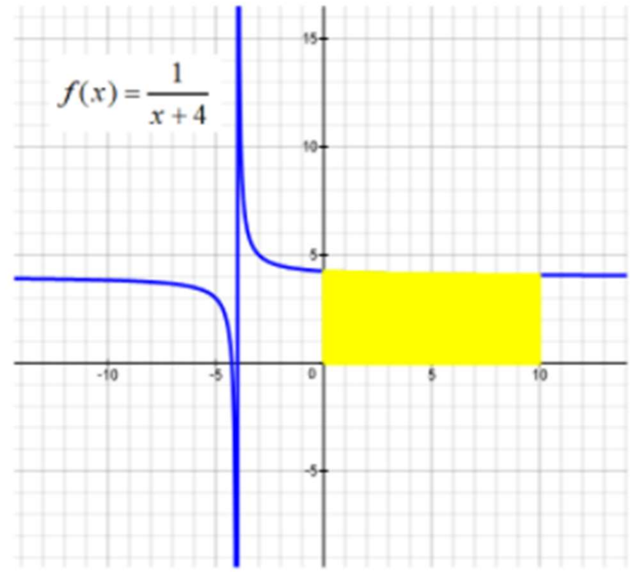


Shaded Area $= \int_a^b f(x) \cdot dx =$ __?__

10) $f(x) = \dfrac{1}{x+4}$ $\quad 0 \le x \le 10$

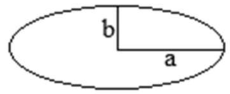
Set up the definite integral representing shaded area.

Shaded Area $= \int_a^b f(x) \cdot dx =$ __?__



11) $f(x) = 4\sqrt{1 - \dfrac{x^2}{25}}$ $\quad 0 \le x \le 5$

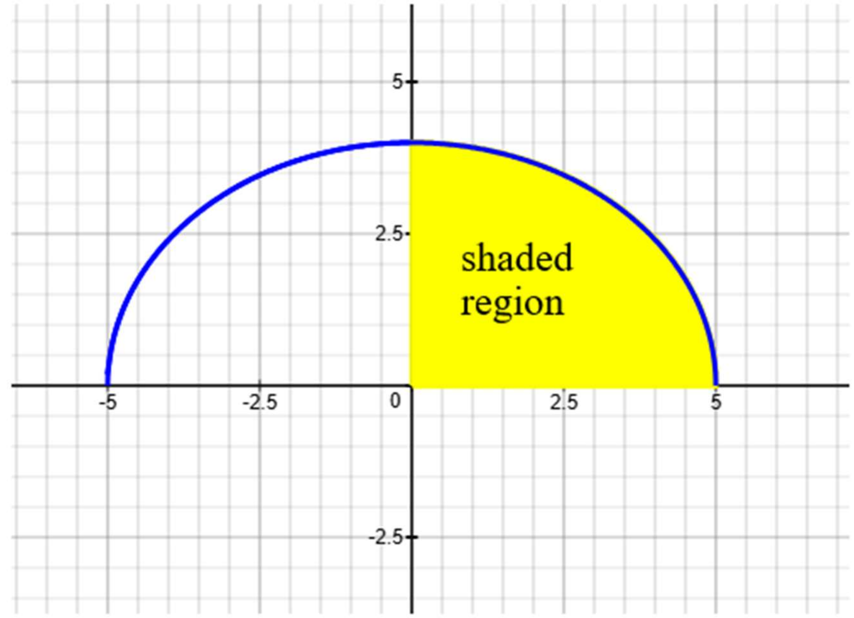
Hint: Area of ellipse = $\pi \cdot a \cdot b$



a) Area of shaded region = __?__

b) Set up the definite integral representing the shaded area.

Shaded Area = $\int_a^b f(x) \cdot dx =$ __?__



12) Set up the definite integrals representing the shaded area

a) Area of Region 1 = $\int_a^b f(x) \cdot dx =$ __?___

b) Area of Region 2 = $\int_c^d g(x) \cdot dx =$ __?___